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Oefeningenreeks 4A nr. 60.14

$$\int \frac{6x + 2 - \sqrt{4x + 1}}{2x + 1 + \sqrt{4x + 1}} dx$$

Neem:

$$t = \sqrt{4x + 1}$$

$$t^2 = 4x + 1$$

$$x = \frac{t^2 - 1}{4}$$

$$dx = \frac{t}{2} dt$$

Dus:

$$\begin{aligned} & \int \frac{6x + 2 - \sqrt{4x + 1}}{2x + 1 + \sqrt{4x + 1}} dx \\ &= \frac{1}{2} \int \frac{t \left(\frac{3(t^2 - 1)}{2} - t + 2 \right)}{\frac{t^2 - 1}{2} + t + 1} dt \\ &= \frac{1}{2} \int \frac{3t^3 - 2t^2 + t}{t^2 + 2t + 1} dt \\ &= \frac{1}{2} \int \left(3t - 8 + \frac{14t + 8}{(t + 1)^2} \right) dt \\ &= \frac{1}{2} \int (3t - 8) dt + \int \frac{7t + 4}{(t + 1)^2} dt \\ &= \frac{3}{4} t^2 - 4t + \int \frac{7t + 4}{(t + 1)^2} dt \\ &= \frac{3}{4} t^2 - 4t + \int \frac{7(t + 1) - 3}{(t + 1)^2} dt \\ &= \frac{3}{4} t^2 - 4t + 7 \int \frac{1}{t + 1} dt - 3 \int \frac{1}{(t + 1)^2} dt \\ &= \frac{3}{4} t^2 - 4t + 7 \ln |t + 1| + \frac{3}{t + 1} + c \end{aligned}$$

Terug substitueren:

$$\begin{aligned} &= \frac{3}{4} (4x + 1) - 4\sqrt{4x + 1} + 7 \ln |\sqrt{4x + 1} + 1| + \frac{3}{\sqrt{4x + 1} + 1} + c \\ &= 3x - 4\sqrt{4x + 1} + 7 \ln |\sqrt{4x + 1} + 1| + \frac{3}{\sqrt{4x + 1} + 1} + c \end{aligned}$$

References